

GRADWAVE: differentiable plane-wave density functional theory in PyTorch

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Abstract

We present GRADWAVE, a plane-wave pseudopotential density functional theory code in which every energy term is a differentiable PyTorch function of its inputs. The code reproduces QUANTUM ESPRESSO 7.5 at the precision established for mature plane-wave codes. Equation-of-state comparisons over six solids give $\Delta \leq 0.17$ meV/atom, lattice constants agree to 4×10^{-4} Å, and a 0.5 ps microcanonical trajectory driven by autograd forces conserves total energy to a secular drift near 1 μ eV/atom. On matched small cells the code runs within a factor of three of QUANTUM ESPRESSO on a consumer GPU. The differentiable formulation then supplies derivatives that conventional codes obtain only from hand-derived, functional-specific implementations. Gradients of the total energy with respect to exchange-correlation parameters match self-consistent finite differences to 3×10^{-10} and support direct fitting of functional parameters. The Hubbard U becomes both a differentiable parameter, through an exact dE/dU , and a self-determined one, through a linear-response implementation whose interacting susceptibility is screened by an exchange-correlation kernel evaluated as an automatic second derivative. The same kernel drives electric-field response, and the resulting dielectric constants and Born charges agree with `ph.x` to 0.002% for Si and to 0.3% or better for MgO. Phonon frequencies from displaced autograd forces agree with density functional perturbation theory to 0.15%. Non-collinear magnetism uses an exchange field generated by the same automatic differentiation, and spin-orbit benchmarks reproduce the GaAs split-off gap to 0.03 meV.

1 Introduction

Plane-wave density functional theory (DFT) rests on a small number of derivative relationships. Forces are derivatives of the total energy with respect to atomic positions, the exchange-correlation (XC) potential is a functional derivative of the XC energy, phonons and dielectric constants are second derivatives, and the Hubbard U of DFT+ U is itself defined through a derivative of occupations with respect to a probe potential [1, 2]. Production codes implement each of these by hand. Every new functional requires a new v_{xc} and a new f_{xc} , every new perturbation requires new matrix elements, and the implementations are validated against one another rather than against the energy expression they differentiate.

Automatic differentiation removes this duplication. If the total energy is a differentiable program, then its gradients with respect to any input, namely positions, XC parameters, occupation-matrix probes, or external fields, are exact by construction and are produced by the same code path that evaluates the energy. Prior work has demonstrated this principle for Hartree-Fock and for basis-set and orbital-free DFT [3, 4]. The question we address is whether a differentiable formulation can meet the precision and performance standards of established plane-wave codes on periodic solids, including metals, magnetism, spin-orbit coupling, and DFT+ U , and what the derivative capabilities provide once it does.

GRADWAVE answers the first part by direct comparison against QUANTUM ESPRESSO 7.5 [5] at identical pseudopotentials, cutoffs, k -meshes, and fast Fourier transform (FFT) grids. Section 3 reports equation-of-state, molecular-dynamics, and response-function benchmarks, and Sec. 4 reports wall-clock comparisons. The second part is addressed in Sec. 5, which collects four capabilities that follow from the computation graph. These are exact gradients with respect to XC

parameters, a differentiable and self-determined Hubbard U , phonons from autograd forces, and noncollinear magnetism with an automatically generated exchange field.

2 Implementation

GRADWAVE implements norm-conserving pseudopotential DFT in the Kleinman-Bylander form with UPF v2 inputs (ONCV and PseudoDojo sets [6, 7, 8]), LDA and PBE functionals [9], Monkhorst-Pack sampling with space-group reduction, Gaussian, Marzari-Vanderbilt, and Methfessel-Paxton smearing, collinear spin, and noncollinear spinors with spin-orbit coupling from fully relativistic j -resolved projectors. All arithmetic is float64 or complex128, on CPU or a single GPU. All results below were obtained on a laptop with 22 hardware threads and an NVIDIA RTX 3050 with 6 GB of memory.

The code is organized so that automatic differentiation never traces the self-consistent field (SCF) loop. A pure tensor layer expresses every energy term and the Hamiltonian apply as differentiable functions. A solver layer runs Davidson diagonalization and Pulay mixing under `torch.no_grad()`. Derivatives are then taken at the converged fixed point in one of three ways. Quantities protected by stationarity, such as forces and $dE/d\theta$ for any energy parameter θ , are partial derivatives at fixed density and orbitals and cost one backward pass. Quantities that require the density response are computed from conduction-projected Sternheimer equations, in which the Hartree-XC kernel f_{Hxc} is applied as a Hessian-vector product of the XC energy by double backpropagation, so no kernel is ever derived by hand. Mixed second derivatives combine the two, with one response solve and one backward pass.

The XC functional is a `torch.nn.Module`. Its potential, its kernel, and its parameter gradients all come from the same energy expression, so a functional with trainable parameters participates in every capability of the code, including the response solvers, without additional implementation.

3 Agreement with Quantum ESPRESSO

The primary validation target is the equation-of-state comparison of Lejaeghere *et al.* [10], in which two

codes are compared through the root-mean-square difference Δ of their fitted energy-volume curves. Total energies were computed at seven volumes spanning $\pm 6\%$ for Si, C, GaAs, MgO, Al, and Cu with PBE, ONCV pseudopotentials, and converged cutoffs of 50 to 90 Ry, and Birch-Murnaghan forms were fitted to both data sets. Figure 1 shows the result. Δ is at or below 0.17 meV/atom for all six solids and at or below 0.01 meV/atom for the four insulators, against a range of 0.3 to 1 meV/atom reported among mature plane-wave codes [10]. Fitted lattice constants agree to 4×10^{-4} Å and bulk moduli to 0.6 GPa. Two numerical conditions were required for smooth curves and are recorded with the benchmark scripts. The FFT grid must be held fixed across volumes, and the cutoff must be converged, since plane-wave shells crossing the cutoff as the cell scales otherwise produce energy jumps that appear identically in both codes.

Energy conservation in molecular dynamics is a stricter test of force-energy consistency than any single-point force comparison, because the trajectory integrates every systematic error. Figure 2 shows 0.5 ps of velocity-Verlet dynamics for a rattled Si_8 cell at an average temperature of 180 K. Potential and kinetic energy exchange roughly 45 meV/atom while the total energy fluctuates with a root-mean-square amplitude of 0.067 meV/atom, the bounded oscillation expected of the integrator at this time step. The secular drift, measured as the difference between the first-half and second-half means, is $0.9 \mu\text{eV/atom}$. The autograd force is the exact gradient of the SCF energy surface, and the drift is at the SCF tolerance floor.

Table 1 collects the remaining benchmark comparisons. All were run at identical numerical settings in both codes, and the full comparison suite of 46 integration tests passes unchanged on CPU and GPU.

4 Performance

Figure 3 compares SCF wall times on seven systems with 1 to 8 atoms, with all numerical settings matched between the codes. QUANTUM ESPRESSO completes these cells in 1.5 to 2.2 s and is faster than the GRADWAVE GPU path on the metals and Si_8 , comparable on the insulators, and 2 to $5\times$ faster than the 22-thread CPU path. Two factors account for most of the difference. QUANTUM ESPRESSO converges in 6 to 9 SCF iterations against 10 to 14 for GRADWAVE, which reflects mixer maturity rather than tensor throughput,

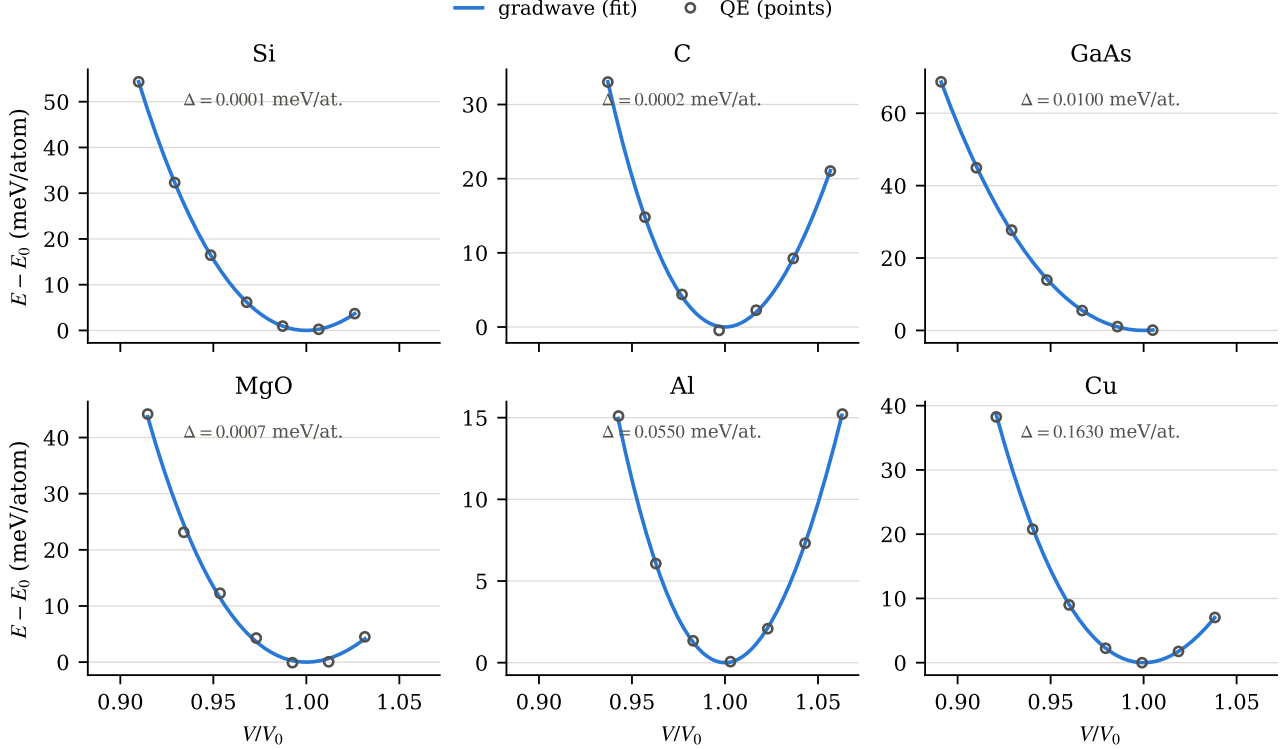


Figure 1: Equations of state for six solids at seven volumes spanning $\pm 6\%$, computed with PBE and ONCV pseudopotentials at identical cutoffs, k -meshes, and FFT grids in both codes. Solid lines show Birch-Murnaghan fits to the GRADWAVE energies and open circles show the QUANTUM ESPRESSO energies. The Δ value in each panel is the root-mean-square difference between the two fitted curves over the volume window [10].

Table 1: Agreement with QUANTUM ESPRESSO 7.5 at identical pseudopotentials, cutoffs, k -meshes, and FFT grids.

quantity	agreement
SCF energies (Si, Al, Cu, Fe, GaAs)	≤ 2 meV/atom
forces, displaced Si and Al	≤ 5 meV/Å
Si band energies	≤ 10 meV
GaAs spin-orbit gap Δ_0	0.3360 vs 0.3360 eV
NiO DFT+ U energy E_U	2.9956 vs 2.9942 eV
ϵ_∞ , Si	23.8900 vs 23.8904
Born charges, MgO	0.07 to 0.10%
Si Γ optical phonon	522.6 vs 523.3 cm^{-1}

and at these sizes, below 3000 plane waves, framework overhead per iteration is proportionally largest. The GPU advantage grows with system size. A 78-electron Bi₂Se₃ spin-orbit calculation on an 80³ grid completes its SCF in under ten minutes on the same 6 GB card. The differentiable formulation therefore remains usable for routine ground-state work.

5 Derivatives from the computation graph

5.1 Exchange-correlation parameters

At self-consistency the total energy is variational in the density, so the gradient with respect to any XC parameter reduces to a partial derivative at fixed density,

$$\frac{dE}{d\theta} = \left. \frac{\partial E_{\text{xc}}[\rho; \theta]}{\partial \theta} \right|_{\rho^*}, \quad (1)$$

which is one backward pass with no response solve. We verified this for a PBE-form exchange functional with learnable (κ, μ) , evaluated at an off-PBE point. The autograd gradients agree with central finite differences of full SCF re-runs to 3×10^{-10} for θ_κ and 7×10^{-10} for θ_μ .

Figure 4 shows the gradients in use. Starting from $(\kappa, \mu) = (0.55, 0.30)$, Adam fits the two parameters to PBE total energies at two silicon volumes. The loss falls three orders of magnitude in five steps, and μ converges to 0.222 against the PBE value 0.2195. κ does

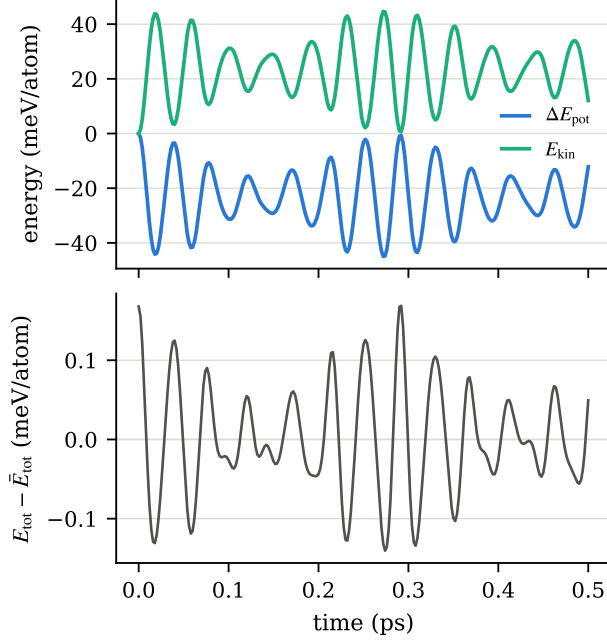


Figure 2: Microcanonical molecular dynamics of a rattled Si_8 cell with a 2 fs time step, driven by autograd Hellmann-Feynman forces through the ASE calculator on the GPU. The upper panel shows the exchange between potential and kinetic energy. The lower panel shows the deviation of the total energy from its mean, with a root-mean-square fluctuation of 0.067 meV/atom and a secular drift near 1 $\mu\text{eV}/\text{atom}$ over the half picosecond.

not converge, and this failure is informative. In silicon’s reduced-gradient range the enhancement factor is $F \approx 1 + \mu s^2$ to leading order, so energy data constrain μ while κ , an s^4 effect, is left nearly free. The optimization performs an identifiability analysis of the functional form on the given data. Losses that depend on the density rather than the energy require the response of the density to the parameters, and this path is implemented as a self-consistent adjoint solve, validated to 2×10^{-4} against finite differences.

5.2 The Hubbard U

GRADWAVE implements rotationally invariant DFT+ U [11] with the atomic-orbital projectors of QUANTUM ESPRESSO, validated on antiferromagnetic NiO. The Hubbard energy agrees with QUANTUM ESPRESSO to 1.6 meV and the Ni 3d occupations to 10^{-3} . Three derivative capabilities follow.

The parameter derivative is exact and costs one back-

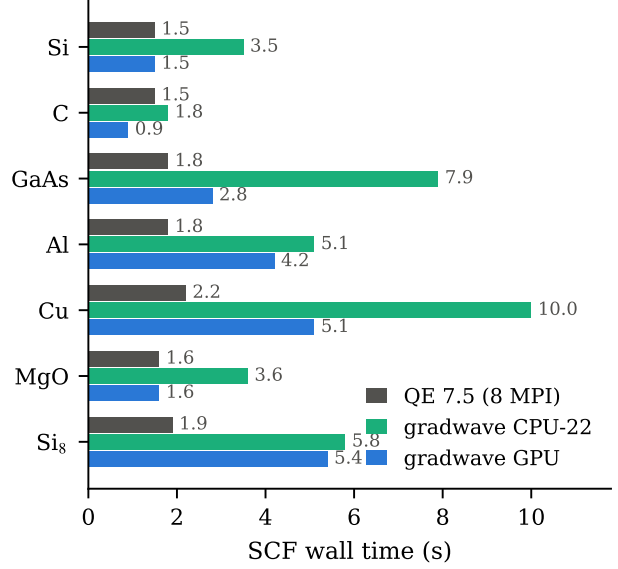


Figure 3: SCF wall time for seven matched systems. QUANTUM ESPRESSO ran with eight MPI ranks, and GRADWAVE ran on 22 CPU threads or on the RTX 3050. Cutoffs, k -meshes, FFT grids, band counts, and functionals are identical, and the converged energies agree to 1 meV/atom or better.

ward pass. At the converged point,

$$\frac{dE}{dU} = \sum_{I\sigma} \frac{1}{2} \text{Tr} [n^{I\sigma} (1 - n^{I\sigma})], \quad (2)$$

by the same stationarity argument as above. For NiO at $U = 5$ eV this gives 0.616154 against 0.616153 from finite differences of SCF re-runs. The force contribution of the Hubbard term follows from backpropagation through the projector phases $e^{-i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}$ and matches frozen-orbital finite differences to 5×10^{-8} eV/Å.

The linear-response U of Cococcioni and de Gironcoli [2] is defined through occupation susceptibilities, $U = (\chi_0^{-1} - \chi^{-1})_{II}$, conventionally evaluated by finite differences of probe SCF calculations. GRADWAVE computes both susceptibilities analytically from a single ground state. The bare χ_0 is one batched conduction-projected Sternheimer solve with the projector probe on the right-hand side. The interacting χ screens the probe self-consistently through the spin-resolved Hartree-XC kernel, which is evaluated as a Hessian-vector product of E_{Hxc} by double backpropagation. No $f_{\text{xc}}^{\sigma\sigma'}$ was derived or coded, so the same solver applies unchanged to any twice-differentiable spin functional, including learnable ones. Figure 5 shows the convergence of the

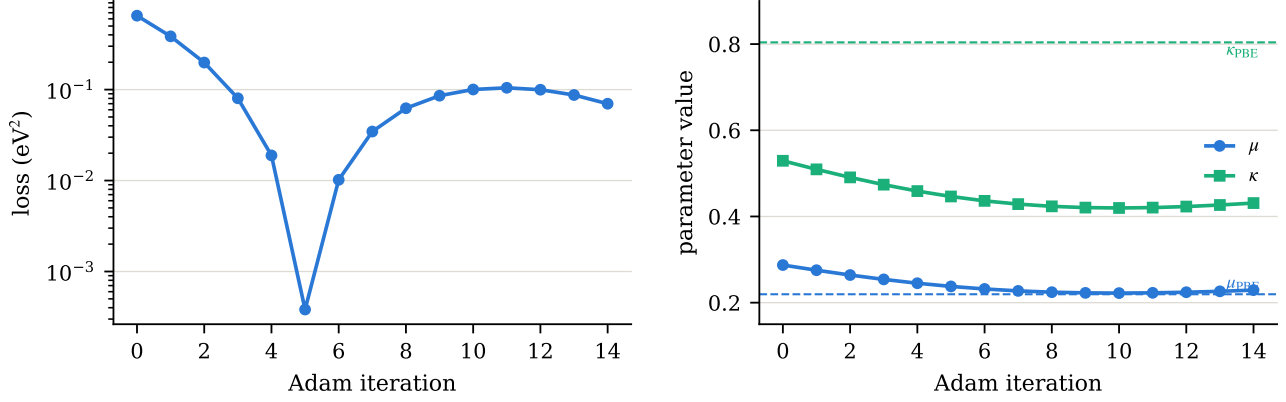


Figure 4: Fit of the learnable exchange parameters (κ, μ) to PBE total energies at two silicon volumes, starting from (0.55, 0.30). The left panel shows the loss and the right panel shows the parameter trajectories, with PBE values as dashed lines. μ converges to 0.222 against the PBE value 0.2195, while κ drifts along a flat direction because silicon’s reduced-gradient range constrains only the s^2 expansion of the enhancement factor.

Table 2: Linear-response Hubbard U for the Ni 3d manifold of NiO (PBE, PseudoDojo, 40 Ry, 2^3 k -mesh).

method	U (eV)
finite-difference response	6.4493
Sternheimer with autograd kernel	6.4496
QUANTUM ESPRESSO <code>hp.x</code>	6.4308

response columns. Anderson mixing [12] is required rather than optional here. PBE NiO lies near a spin instability, the antisymmetric magnetization mode across the two Ni sites gives the iteration matrix an eigenvalue near -6 , and plain damped iteration diverges at any practical damping. Anderson mixing converges the same fixed point in 11 iterations.

Table 2 compares the resulting U against the finite-difference route and against the independent density functional perturbation theory implementation in `hp.x` [13]. The two GRADWAVE routes agree to 0.3 meV, and both agree with `hp.x` to 0.3%.

5.3 Phonons and electric-field response

Phonons were computed by finite displacements of 5×10^{-3} Å with autograd forces, at settings matched to a `ph.x` calculation (PBE, 40 Ry, 4^3 mesh, 32^3 grid). The silicon Γ optical triplet is 522.6 cm^{-1} against 523.3 cm^{-1} from perturbation theory, a 0.15% difference, with the triplet degenerate to the reported precision and the acoustic modes at zero under the sum rule. Because

the forces are exact gradients, the force-constant matrix is consistent with the energy surface by construction, and the corresponding diagonal Hessian element agrees with an energy second difference to 0.5%.

The electric-field response uses the same Sternheimer and autograd-kernel machinery as the Hubbard U . The position operator acting on Bloch states is obtained from a solve with $\partial H / \partial k$ on the right-hand side, where the kinetic contribution is analytic and the nonlocal contribution is a central difference in k of the projector tables. The result is insensitive to the step over a factor-of-four range at the sixth decimal. The dielectric constant of silicon agrees with `ph.x` to 0.002%. Born effective charges follow as mixed derivatives $\partial^2 E / \partial \mathcal{E} \partial \tau$, in which one backward pass through the position-differentiable pseudopotential yields all atoms and directions per field direction. For MgO the charges agree with `ph.x` to 0.07% for Mg and 0.10% for O, and the raw acoustic sum rule violation, a k -mesh artifact that both codes must reproduce before enforcement, matches as well.

5.4 Noncollinear magnetism

The noncollinear SCF treats the exchange field the same way the collinear code treats the potential. For a locally collinear functional the energy depends on $(\rho, |\mathbf{m}|)$, and a single backward pass yields both v_{xc} and $\mathbf{B}_{xc} \parallel \mathbf{m}$, so no functional derivative of the spin rotation was implemented. The consequences are testable symmetries rather than fitted numbers. Spinor iron with $\mathbf{m} \parallel \hat{z}$ re-

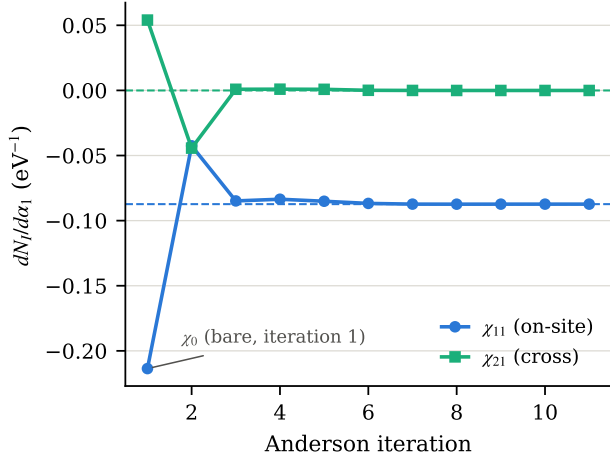


Figure 5: Convergence of the interacting response column $dN_I/d\alpha_1$ for NiO under Anderson mixing. The first iteration is the bare response χ_0 , computed at frozen effective potential. Dashed lines mark the finite-difference reference values obtained from probe SCF reruns. Plain damped iteration diverges for this system, because the antisymmetric magnetization mode across the two Ni sites carries a $K\chi_0$ eigenvalue near -6 .

produces the two-channel collinear energy exactly at the reported precision, and rotating the moment to $\hat{x}$ or to 45° changes the total energy by $0.19 \mu\text{eV}$ in the absence of spin-orbit coupling, so the rotational invariance of the functional emerges numerically from the automatic field.

With fully relativistic j -resolved projectors the code reproduces standard spin-orbit benchmarks. The GaAs split-off gap matches QUANTUM ESPRESSO to 0.03 meV and the free energy to 0.0007 meV/atom . For Bi_2Se_3 the Γ -point parities of the gap-edge states swap between the scalar-relativistic and spin-orbit calculations, with parity eigenvalues of ± 1.000 and exact Kramers degeneracy, which is the band inversion underlying its topological classification. Magnetic energy differences are also quantitative. Chromium orders antiferromagnetically at -27.7 meV/atom relative to the nonmagnetic state, matching QUANTUM ESPRESSO to 0.02 meV/atom , and mapping iron energy differences onto a Heisenberg model gives $J_1 = 57.5 \text{ meV}$ and $J_2 = -39.6 \text{ meV}$.

6 Discussion

The comparisons above support two conclusions. A differentiable plane-wave DFT code can match a reference implementation at the precision mature codes achieve among themselves, across ground states, forces, response functions, magnetism, and spin-orbit coupling. And once the energy is a differentiable program, a family of derived quantities stops requiring separate implementations. The XC kernel used by every response solver in this work is a second derivative of the same energy expression that the SCF minimizes, the Born charges are one backward pass, and the Hubbard U gradient is a trace of quantities the $+U$ SCF already computes. Each of these normally exists in a production code as its own hand-derived module, specialized to a particular functional.

The present limitations are explicit. The response solvers cover insulators, since the metallic Fermi-surface response term is not yet implemented. Stress and variable-cell relaxation require radial tables that are differentiable in the cell, and the SCF mixer costs roughly $1.5\times$ the iterations of QUANTUM ESPRESSO's. None of these is structural. The most direct next steps are the metallic response term, LO-TO splitting from the computed Born charges, and magnetocrystalline anisotropy torques $dE/d\hat{\mathbf{m}}$, for which all ingredients are in place.

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